

Database Project

Food Delivering Partner DBMS

Course	IT214 – Database Management System, Autumn'2026
Instructor	PM Jat
Domain	Food Delivery, Restaurant Discovery & Grocery Delivery Platform
Inspired By	Swiggy, Zomato

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1. Objective

Design and implement a database for an online food delivery, restaurant discovery, and grocery delivery platform. The platform serves two core verticals:

- Food Delivery – Connects customers with restaurants in their area, enabling them to browse menus, place food orders, and get them delivered to their doorstep via delivery partners.
- Grocery Delivery – Connects customers with grocery/dark stores and supermarkets for quick delivery of daily essentials, fruits, vegetables, dairy, and packaged goods.

Let the platform be called QuickBite.in

The database must support the entire lifecycle of both order types: from a customer discovering a restaurant or grocery store, browsing items, placing an order, status tracking, delivery by a delivery partner, payment processing, and post-delivery ratings and reviews.

2. Application Users

QuickBite.in has the following categories of users:

- Surfer – Unregistered/Guest users who browse without logging in
 - Customers (Buyers) – Registered users who order food or groceries
 - Restaurant Partners – Registered restaurants that list menus and receive food orders
 - Grocery Store Partners – Registered grocery stores/dark stores that list product inventory and receive grocery orders
 - Delivery Partners – Registered delivery personnel who pick up and deliver both food and grocery orders
 - Platform Admin – QuickBite.in system administrators
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3. Use Cases

3.1 Use Cases – Surfers (Unregistered Users)

These users explore the platform without registering.

1. Search restaurants by name, cuisine type, location/area, or food item name
2. Browse restaurant listings with basic info: name, cuisine, average rating, delivery time estimate, minimum order value
3. View a restaurant's menu with item names, descriptions, prices, and veg/non-veg tags
4. Search grocery stores by name or location
5. Browse grocery store product catalog by category (e.g., Fruits & Vegetables, Dairy, Snacks, Beverages)
6. View product details: name, brand, quantity/weight, price, availability
7. View customer reviews and ratings for a restaurant or grocery store

3.2 Use Cases – Customers (Buyers)

These are registered users who order food or groceries through the platform.

Registration and Profile Management

8. Register on the platform by providing: Full Name, Email, Phone Number, and Password
9. Log in and log out of the account
10. Update personal profile details
11. Manage multiple saved delivery addresses (Home, Work, Other) with label, address line, city, pincode, and GPS coordinates

Restaurant and Menu Browsing

1. Search and filter restaurants based on:
 - Location / delivery area
 - Cuisine type (e.g., Indian, Chinese, Italian, Fast Food)
 - Dietary preference (Pure Veg / Non-Veg)
 - Minimum order amount
 - Average rating
 - Delivery time
12. View detailed restaurant page: name, address, cuisine, timings, contact, average rating, photos
13. Browse restaurant menu categorized by sections (e.g., Starters, Mains, Beverages, Desserts)

14. View individual menu item details: name, description, price, portion size, veg/non-veg flag, availability status, and item ratings

Grocery Store Browsing and Shopping

15. Search and browse grocery stores available in the customer's delivery area
16. Browse grocery product catalog by category (Fruits & Vegetables, Dairy & Eggs, Bakery, Snacks, Beverages, Personal Care, Household, etc.)
17. Search for a specific product by name or brand across all available grocery stores
18. View product details: name, brand, description, unit/weight/quantity, MRP, selling price, discount percentage, availability, and expiry date category
19. View and compare prices of the same product available across different grocery stores

Cart and Order Management

20. Add food items or grocery products to cart (cart is scoped to a single store/restaurant at a time)
21. View, edit (change quantity), or remove items from cart
22. Apply coupon/promo codes for discounts on food or grocery orders
23. Place an order by selecting a saved delivery address and payment method
24. Track order status in real-time. For food orders: Placed, Confirmed, Preparing, Out for Delivery, Delivered. For grocery orders: Placed, Confirmed, Packed, Out for Delivery, Delivered
25. View full order history for both food and grocery orders
26. Cancel an order before it is confirmed/accepted
27. Raise complaints or refund requests for issues with a delivered order

Reviews and Ratings

28. Rate a restaurant or grocery store (scale 1–5) after order delivery
29. Rate individual food items or grocery products ordered (scale 1–5)
30. Rate the delivery partner (scale 1–5)
31. Write review comments for a restaurant, grocery store, or a product

Favorites and Wishlist

32. Mark restaurants or grocery stores as favorites for quick access
33. Save individual food items or grocery products to a personal wishlist

3.3 Use Cases – Restaurant Partners

Restaurants register as partners to receive and fulfill orders.

Registration and Profile Management

34. Register on the platform by providing: Restaurant Name, Owner Name, FSSAI License Number, PAN/GSTIN, Bank Account Details (for settlement), Restaurant Address with GPS coordinates, Contact Number, Cuisine Types, and Operating Hours
35. Upload restaurant photos and logo
36. Set/update restaurant operating hours and mark holidays/off days
37. Temporarily mark restaurant as closed (e.g., during rush hours or emergencies)

Menu Management

38. Create and manage menu categories (e.g., Starters, Main Course, Combos, Beverages)

- 39. Add new menu items with: Item Name, Description, Price, Category, Veg/Non-Veg flag, Availability, and optional photo
- 40. Update item prices and availability (mark items as out-of-stock temporarily)
- 41. Delete or deactivate menu items
- 42. Create combo/meal deals linking multiple items at a special bundled price

Order Management

- 43. Receive incoming order notifications with full order details
- 44. Accept or reject an incoming order (with rejection reason)
- 45. Update order status: Confirmed -> Preparing -> Ready for Pickup
- 46. View list of active/pending orders

Reports and Analytics

- 47. View daily/weekly/monthly order and revenue summaries
- 48. View item-wise sales report (most ordered items, revenue per item)
- 49. View customer reviews and ratings received
- 50. View pending settlement amounts from QuickBite.in

3.4 Use Cases – Grocery Store Partners

Grocery stores and dark stores register as partners to list products and fulfill grocery orders.

Registration and Profile Management

- 51. Register by providing: Store Name, Owner Name, FSSAI License, GSTIN, Bank Account Details, Store Address with GPS coordinates, Contact Number, and Operating Hours
- 52. Specify store type: Supermarket / Dark Store / Pharmacy / Specialty Store
- 53. Set delivery radius and estimated delivery time (e.g., 10 minutes, 30 minutes)
- 54. Temporarily mark store as closed or pause accepting new orders

Inventory and Product Management

- 55. Create and manage product categories (e.g., Fruits & Vegetables, Dairy & Eggs, Snacks, Beverages, Personal Care, Household Supplies)
- 56. Add new products with: Product Name, Brand, Description, Category, Unit Type (kg/litre/piece/pack), Net Weight/Volume, MRP, Selling Price, Stock Quantity, Veg/Non-Veg/Vegan flag, Expiry Type (perishable/non-perishable), and product image
- 57. Update product prices, discounts, and stock quantities
- 58. Mark products as out-of-stock or discontinue them
- 59. Manage product variants (e.g., 500ml, 1 litre, 2 litre for same product)

Order Management

- 60. Receive incoming grocery order notifications
- 61. Accept or reject an order (with rejection reason, e.g., item unavailable)
- 62. Update order status: Confirmed -> Packing -> Ready for Pickup
- 63. Substitute an unavailable item with a similar product (with customer notification)
- 64. View list of active/pending orders

Reports

- 65. Daily/weekly/monthly order count and revenue summary
- 66. Product-wise sales report: most sold items, low stock alerts
- 67. Customer reviews and ratings received

68. Pending settlement amounts from QuickBite.in

3.5 Use Cases – Delivery Partners

These are registered delivery personnel who pick up orders from restaurants and deliver them to customers.

- 69. Register with: Full Name, Phone Number, Vehicle Type (Bicycle/Motorcycle/Car), Vehicle Registration Number, Driving License Number, Bank Account for salary/incentive credit
- 70. Set availability status (Online/Offline)
- 71. Receive order delivery assignment notifications
- 72. Accept or decline a delivery assignment
- 73. Update delivery status: Picked Up -> On the Way -> Delivered
- 74. View delivery history and earnings summary

3.6 Use Cases – Platform Admin

- 75. Approve or reject restaurant and delivery partner registrations
- 76. Manage platform-wide coupon/promo codes (create, edit, deactivate)
- 77. View and manage customer complaints and refund requests
- 78. View platform-wide sales, order, and revenue dashboards
- 79. Manage commission rates for restaurants
- 80. View and manage financial settlements with restaurant partners

4. Key Data Requirements

4.1 Users (Customers, Delivery Partners)

- UserID (unique across all users), Full Name, Email, Phone, Password Hash
- Role: Customer / Delivery Partner / Restaurant Owner / Admin
- Registration date, Last login timestamp, Account status (Active/Suspended)
- For Customers: multiple saved delivery addresses (label, address, city, pincode, GPS lat-long)
- For Delivery Partners: vehicle type, vehicle number, license number, current availability status, current GPS location (last known)

4.2 Restaurants

- RestaurantID, Restaurant Name, Owner details, FSSAI License, GSTIN
- Address, City, Area/Zone, GPS coordinates (lat-long)
- Cuisine types (a restaurant can serve multiple cuisines)
- Operating hours per day of week, current open/closed status
- Minimum order amount, average delivery time (in minutes), delivery radius (in km)
- Average rating (computed from reviews), total number of ratings
- Bank account details for settlement, Commission percentage agreed with platform
- Registration status: Pending / Approved / Suspended

4.3 Grocery Stores

- StoreID, Store Name, Owner details, FSSAI License, GSTIN, Store Type (Supermarket/Dark Store/Pharmacy/Specialty)
- Address, City, Area/Zone, GPS coordinates (lat-long)
- Operating hours, current open/closed status
- Delivery radius (km), estimated delivery time (minutes), minimum order amount
- Average rating, total ratings count
- Bank account details, commission percentage
- Registration status: Pending / Approved / Suspended

4.4 Product Catalog (Grocery)

- Product Categories: CategoryID, StoreID, Category Name, Parent Category (for hierarchical categories), display order
- Products: ProductID, StoreID, CategoryID, Product Name, Brand, Description, Unit Type, Net Weight/Volume, MRP, Selling Price, Stock Quantity, Veg/Non-Veg/Vegan flag, Is Perishable flag, Product Image URL, Average Rating
- Product Variants: VariantID, ProductID, Variant Label (e.g., 500ml / 1L), MRP, Selling Price, Stock Quantity

4.5 Menu and Items (Food)

- Menu Categories: CategoryID, RestaurantID, Category Name, display order
- Menu Items: ItemID, RestaurantID, CategoryID, Item Name, Description, Price, Veg/Non-Veg flag, Availability, photo URL, average item rating
- Combo Deals: ComboID, Name, Price, list of included ItemIDs

4.6 Orders

- OrderID, CustomerID, Order Type (FOOD / GROCERY), RestaurantID or StoreID (depending on type), DeliveryPartnerID
- Order Timestamp, Status — Food orders: Placed/Confirmed/Preparing/Ready/Out-for-Delivery/Delivered/Cancelled. Grocery orders: Placed/Confirmed/Packing/Ready/Out-for-Delivery/Delivered/Cancelled
- Delivery Address snapshot, Estimated Delivery Time
- List of ordered items: ItemID or ProductID, Quantity, Unit Price at time of order, Subtotal
- For grocery orders: item substitution info (original item, substituted item, customer approved flag)
- Pricing breakdown: Items Total, Delivery Fee, Discount (coupon), Taxes, Grand Total
- Payment: Method (UPI/Card/Cash/Wallet), Payment Reference ID, Payment Status
- Cancellation details: Cancelled by, Reason, Timestamp
- Refund details: Refund Amount, Status, Reference ID, Timestamp

4.7 Ratings and Reviews

- ReviewID, OrderID, CustomerID, RestaurantID, Rating (1–5), Review Text, Timestamp
- Item-level ratings: ItemRatingID, OrderID, ItemID, CustomerID, Rating (1–5)
- Delivery Partner ratings: PartnerRatingID, OrderID, DeliveryPartnerID, CustomerID, Rating (1–5), Comments

4.8 Coupons and Promotions

- CouponID, Code, Discount Type (Percentage/Flat), Discount Value, Minimum Order Value, Max Discount Cap

- Applicable Order Type: Food / Grocery / Both
- Applicable to: All Stores / Specific Restaurant(s) / Specific Grocery Store(s)
- Valid From, Valid Until, Max Usage Limit, Times Used

4.9 Financial Settlements

- SettlementID, Partner Type (Restaurant / Grocery Store), PartnerID, Period (From Date – To Date)
- Total Orders, Gross Order Value, Platform Commission Amount, Net Payable to Partner
- Settlement Status (Pending / Processed), Settlement Date, Payment Reference

5. Tentative List of Queries / Reports

For Surfers / General Users

81. Find restaurants within a delivery area serving a specific cuisine
82. Find restaurants serving a specific food item
83. Compare restaurants by rating, delivery time, and price for a given area
84. View complete menu with all categories and items for a restaurant
85. Find grocery stores available in a delivery area
86. Search for a specific grocery product and compare prices across stores
87. Browse grocery store catalog by product category

For Customers [CUSTOMER-QUERIES]

88. My active order (food or grocery) and its current status
89. My full order history – food orders and grocery orders separately
90. My saved addresses
91. My refund status for a disputed order
92. My favorite restaurants and grocery stores
93. Restaurants and grocery stores near me that are currently open
94. Top-rated restaurants in my area by cuisine
95. Grocery products on my wishlist and their current prices

For Restaurant Partners [RESTAURANT-QUERIES]

96. Pending orders to be confirmed
97. Orders ready for delivery pickup
98. Daily/weekly/monthly order count and revenue summary
99. Item-wise sales report: most ordered items, revenue per item
100. Customer ratings and review comments received
101. Menu items that are out of stock
102. Settlement amount pending from QuickBite.in

For Grocery Store Partners [STORE-QUERIES]

103. Pending grocery orders to be confirmed and packed
104. Orders ready for delivery pickup
105. Daily/weekly/monthly order and revenue summary
106. Product-wise sales report: most sold items and revenue per product

- 107. Low stock alerts: products with stock quantity below threshold
- 108. Items that were frequently out-of-stock or substituted in orders
- 109. Customer ratings and reviews received for the store
- 110. Settlement amount pending from QuickBite.in

For Delivery Partners [DELIVERY-QUERIES]

- 111. My assigned active delivery order (food or grocery)
- 112. My delivery history for today/this week
- 113. My total earnings this week/month
- 114. My average rating from customers

For Admin [ADMIN-QUERIES]

- 115. Platform-wide daily/weekly/monthly order and revenue summary (food vs grocery breakdown)
- 116. City-wise, area-wise, cuisine-wise order breakdown for food orders
- 117. Category-wise order breakdown for grocery orders
- 118. Restaurant-wise and grocery store-wise revenue and commission earned
- 119. Top-rated and most-ordered-from restaurants and grocery stores
- 120. Most popular food items and grocery products platform-wide
- 121. Pending restaurant and grocery store registration approvals
- 122. List of all active complaints and refund requests
- 123. Pending financial settlements with restaurant and grocery store partners
- 124. Active coupons and their usage statistics (food vs grocery)
- 125. Delivery partners currently online
- 126. Customers who placed the most orders (food and grocery combined)

6. Scope and Boundaries

The following are explicitly within scope of the QuickBite.in database:

- Restaurant discovery, menu browsing, and food ordering
- Grocery store discovery, product catalog browsing, and grocery ordering
- Product inventory and stock management for grocery stores
- Order lifecycle management from placement to delivery (both food and grocery)
- Delivery partner assignment and status tracking
- Ratings and reviews for restaurants, grocery stores, food items, grocery products, and delivery partners
- Coupon/promo code management (food and grocery)
- Financial settlement records between platform and all partner types
- Complaint and refund request tracking

The following are explicitly out of scope:

- Real-time GPS tracking of delivery partners (only last known location stored)
- Live payment gateway processing (only payment reference and status stored)
- Push notification infrastructure
- Machine learning based recommendation engine (only query-based filtering)

- Cold-chain / temperature tracking for perishable grocery items
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7. Summary of Key Entities

Based on the use cases and data requirements, the following primary entities are identified for the ER Diagram:

- USER (supertype) → CUSTOMER, DELIVERY_PARTNER, STORE_OWNER, RESTAURANT_OWNER, ADMIN
- RESTAURANT
- MENU_CATEGORY
- MENU_ITEM
- COMBO_DEAL
- GROCERY_STORE
- PRODUCT_CATEGORY
- PRODUCT
- PRODUCT_VARIANT
- CUSTOMER_ADDRESS
- CART / CART_ITEM
- ORDER / ORDER_ITEM (covers both food and grocery)
- ORDER_SUBSTITUTION (grocery-specific)
- PAYMENT
- CANCELLATION
- REFUND
- DELIVERY
- RATING_REVIEW (Restaurant, Grocery Store, Food Item, Grocery Product, Delivery Partner)
- COUPON
- SETTLEMENT
- CUISINE (reference table)
- AREA / ZONE (reference table)